21.5 Nondeterministic Automata

The finite state automata we have seen so far are often called **deterministic finite-state automata** or DFAs. There is a generalization called a **non-deterministic finite-state automaton** or NFA. The only difference is how the transition function is specified. For an NFA, the transition function has the form

$$f : S \times I \to P(S)$$

where S is the set of states, I is the input alphabet, and $P(S)$ is the powerset of S. (The powerset of S is the set of all subsets of S.) So the transition function for an NFA returns a *set* of states. That set could contain just one state, or multiple states, or no states at all (empty set).

Here is a tabular description of the transition function for an NFA N_0 with start state A and accepting states C and D.

N_0								
state	A	A	B	B	C	C	D	D
letter	0	1	0	1	0	1	0	1
f	$\{A, B\}$	$\{D\}$	$\{A\}$	$\{B, D\}$	$\{\}$	$\{A, C\}$	$\{A, B, C\}$	$\{B\}$

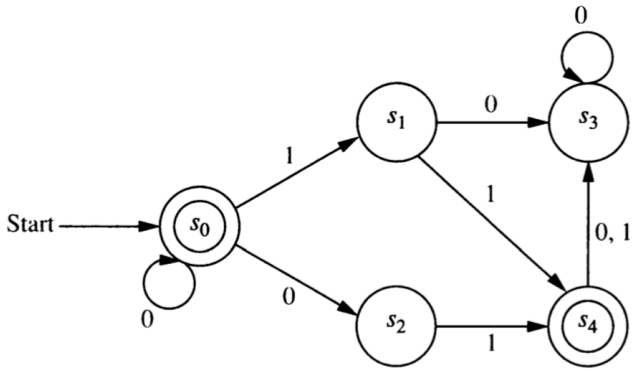
Exercise 21.10. Draw the graph representation. How will the graph of an NFA differ from the graph for a DFA?

Exercise 21.11. How do we define the language recognized by an NFA? [Check your answer before proceeding.]

Exercise 21.12. Which of these strings are accepted by N_0?

a. 01
b. 011
c. 00
d. 0000
e. λ
f. 010101

Here is the graph of an NFA N_1.



Exercise 21.13. Make a table showing the transition function for N_1.

Exercise 21.14. Which of the following strings are accepted by N_1?

a. λ
b. 00
c. 01
d. 010
e. 011
f. 001

Exercise 21.15. Which is harder: to convince someone that a string *is* accepted by an NFA or that it *is not* accepted by an NFA? Why?

Exercise 21.16. What is $L(N_1)$?

Exercise 21.17. For each of the following languages

- Construct an NFA that recognizes the language.
- Construct a DFA that recognizes the language.

In each case, try to do this with as few states and transitionas as you can. You may find it helpful to give your states good names that reflect what they represent.

a. The set of bitstrings that either start 01 or start 10.
b. The set of bitstrings that end with two 0's
c. The set of bitstrings that either end 01 or end 10.

Which do you find easier to construct: NFAs or DFAs? Why?

Exercise 21.18. Create a DFA that recognizes $L(N_1)$.

Exercise 21.19. Given an NFA N, is it always possible to create a DFA M that recognizes the same language? If so, explain how. If not, provide an example N and explain why $L(N)$ cannot be recognized by a DFA.

21.6 The Equivalence of DFA and NFA.

Despite their apparent differences, DFAs and NFAs recognize exactly the same set of languages. That is

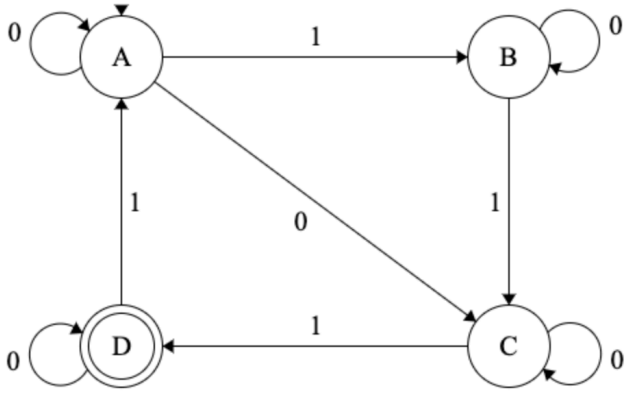
- If the language L is recognized by a DFA M, then there is also and NFA N that recognizes L.
- If the language L is recognized by a DFA N, then there is also and NFA M that recognizes L.

To prove this, we need

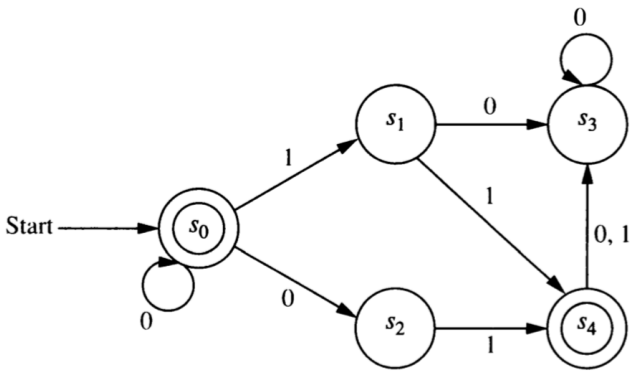
- Algorithm 1: an algorithm that converts DFAs into equivalent NFAs
- Algorithm 2: an algorithm that converst NFAs into equivalent DFAs.

Exercise 21.20. One of these two algorithms is super easy. Which one? What is the algorithm?

Exercise 21.21. The other direction is more interesting. Let's see if we can figure out a general algorithm for this task by first exploring an example. Consider the following NFA. Create an equivalent DFA, using a method that could be applied to any NFA.



Exercise 21.22. Now convert the NFA N_1 into an equivalent DFA. Here's N_1 again.



Exercise 21.23. If an NFA has n states and we use this method to convert it to a DFA, what is the most states the DFA might have?